

Particulate Matter

Monitoring & Health Effects

Issue 2026-09-30 | Entry window 30 September 2026 (full day) | 25 records screened in scope

25RECORDS IN SCOPE
(85 EXCLUDED)**9**OF 10 SUBTOPIC
BANDS POPULATED**9**EFFECT ESTIMATES
WITH A CI**6**TIER-A
RECORDS**8**RECORDS CARRIED ON
METADATA ALONE

Signal of the day

Per microgram, wildfire-smoke carbon is the most lethal fraction of $PM_{2.5}$ in 77.5 million Medicare beneficiaries — and a mass-only network cannot see it.

Jin and colleagues (*ES&T*, tier A) linked annual high-resolution smoke and non-smoke $PM_{2.5}$, split into mass and carbonaceous matter, to Medicare enrollees 2002–2019. Per $1 \mu g m^{-3}$, all-cause mortality HR was **1.032** (1.027–1.037) for smoke $PM_{2.5}$ and **1.087** (1.076–1.098) for smoke carbon — both above their non-smoke counterparts. On the IQR scale the mass ranking **reverses**, because chronic smoke exposure varies little between places; the carbon fraction stays the stronger predictor either way. The authors attribute **22,420 deaths a year** to smoke $PM_{2.5}$, most of it to its carbon, with larger effects in Black and dual-eligible beneficiaries. **Caveats:** annual means flatten episodic smoke into a chronic metric, and the smoke/non-smoke split is itself modelled. **Why it matters for sensing:** optical LCS report mass, and their response to smoke differs from their response to urban aerosol. In fire regions a network that adds an absorption or BC channel measures the fraction that drives the mortality estimate, not just its mass.

What changed today

- **Half of global tuberculosis attributed to $PM_{2.5}$** (*Ann Am Thorac Soc*, 140 studies, tier A): household solid fuel RR **1.82** (1.52–2.17), outdoor short-term $PM_{2.5}$ RR **1.03** (1.00–1.05) per $10 \mu g m^{-3}$, giving **4.15 million** cases and **~535,000** deaths in 2023 (49.9%). The fraction rests on poverty-confounded fuel contrasts; read it as an upper bound.
- **Acute $PM_{2.5}$ and cancer deaths, Japan** (47 prefectures, 2013–2022): **+0.88 %** (0.57–1.18) per $10 \mu g m^{-3}$ at lag 0–2, gone after NO_2 or SO_2 adjustment. **MS relapses, Paris** (1,152 patients, self-controlled): PM_{10} IRR **1.28** and $PM_{2.5}$ **1.20** at a two-week lag; the 11–12-week hits are likely multiplicity.
- **Meteorological adjustment.** A preprint covering ten French cities finds boundary-layer height carries **nine times** the unique variance of wind speed for PM (four times for NO_2 and O_3), despite $r \approx 0.7$ between them — trend normalisation and short-term models that adjust for wind alone are misspecified. A mobile-monitoring BC model for a Beijing pregnancy cohort reaches CV R^2 **0.69** but is not validated at the homes it is used for.
- **Policy modelling.** A Bayesian inversion shows emission cuts of similar tonnage range from full compliance to almost none with the $9 \mu g m^{-3}$ NAAQS, depending on where and what is cut; maximal afforestation in China adds only **$0.39 \mu g m^{-3}$** $PM_{2.5}$ in summer, and two-thirds less under carbon-neutral emissions.
- **Eight of 25 records are metadata-only (32 %)**, seven of them Elsevier deposits, including an extreme-condition **sensor test chamber** (*Atmos Environ*) and **satellite $PM_{2.5}$ composition** for Seoul, both first on the retry list. Scholar Gateway, the full-text route for these, is still disconnected.

Window continuity

The 29 September issue closed with `last_entry_date` at 29 September. This issue collects **30 September in full**, continuous with it. The run started at 17:36 CDT on 1 October, before the 22:00 cut, so the newest buildable date is 30 September and **1 October is left to the next run**. The **September monthly** is built on this run, after this issue, so it includes it.

Corpus analytics

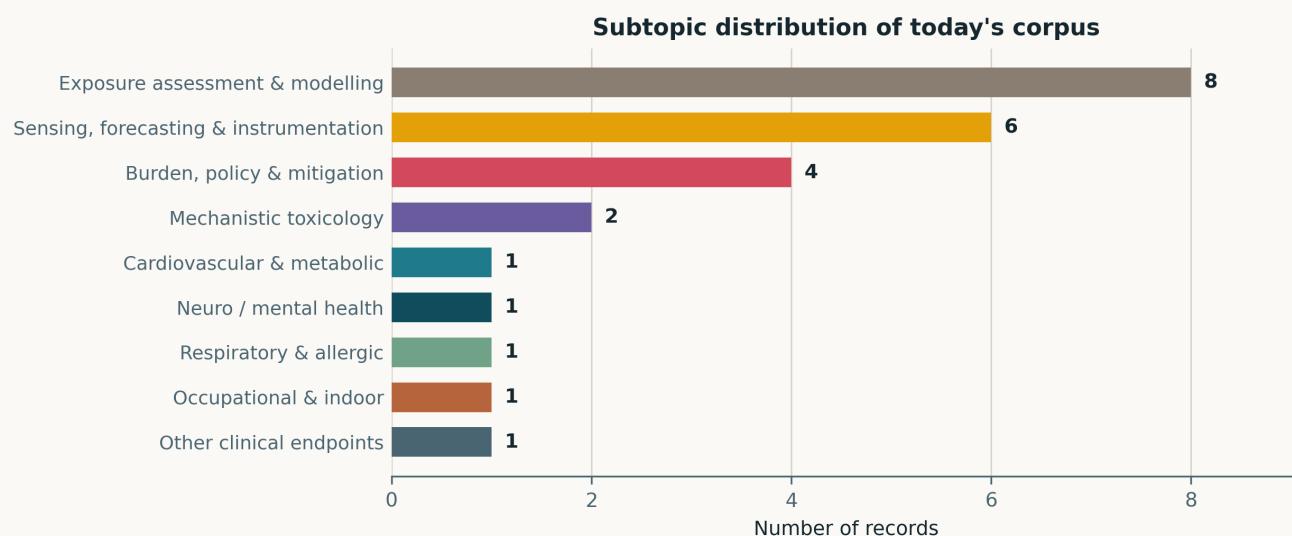


Fig. Subtopic assignment of the 25 in-scope records. **Nine of ten bands populated**; *Exposure assessment* leads with **eight** (five of them metadata-only), *Sensing* has six and *Burden & policy* four. *Reproductive & developmental* is empty; the one prenatal record is a mouse study filed under mechanistic toxicology.

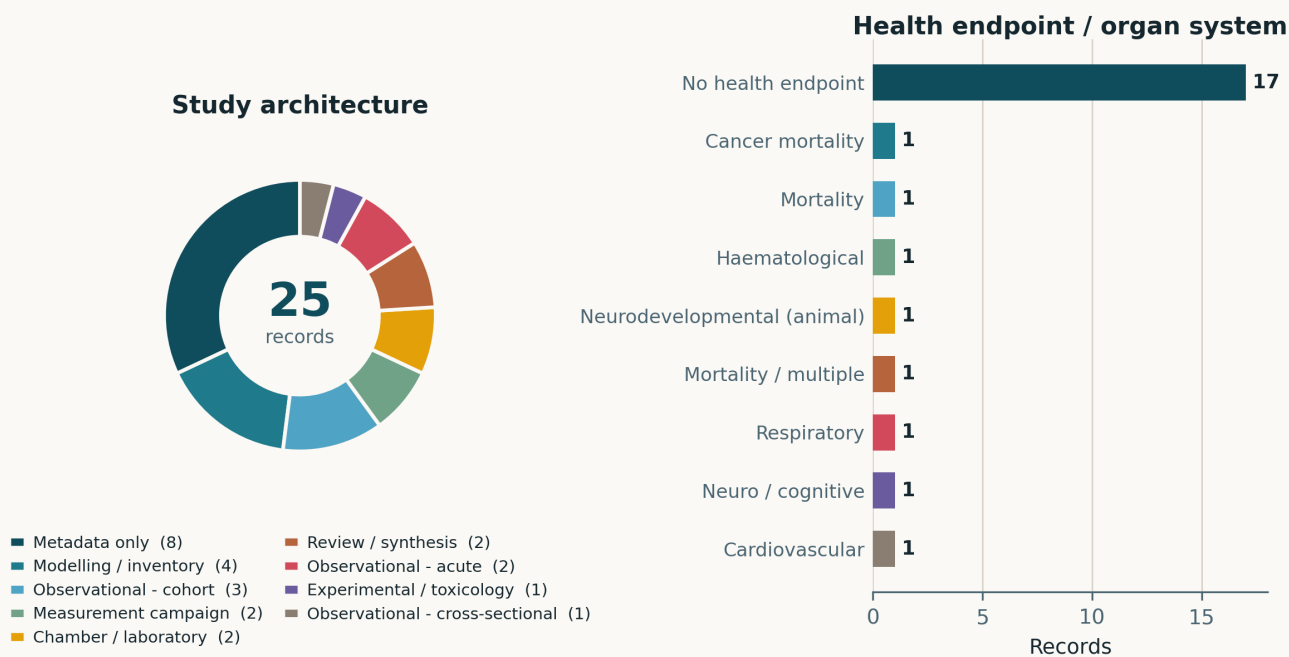


Fig. Left — architecture: metadata-only **eight**, modelling four, cohort three, and two each for measurement campaigns, chamber/laboratory, reviews and acute designs (the case-crossover and the self-controlled MS series); one animal and one cross-sectional study. Right — **17** records carry no health endpoint; the eight that do are spread one each across neurological, respiratory, cardiovascular, cancer mortality, all-cause mortality, multi-outcome proteomics, haematological and animal neurodevelopment.

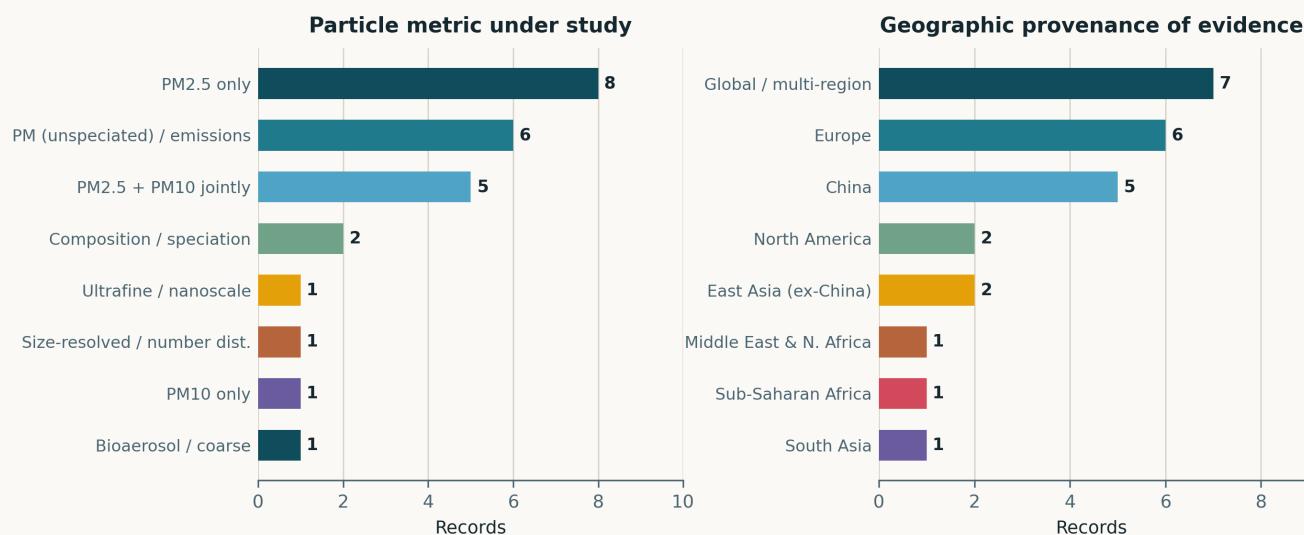


Fig. Left — particle metric: PM_{2.5} only eight, unspeciated / emissions six, PM_{2.5} + PM₁₀ five, composition two, one each for PM₁₀ only, bioaerosol, size-resolved and ultrafine. Right — geography: global / not stated seven (the metadata-only records dominate), Europe six, China five, East Asia ex-China and North America two each, and one each for South Asia, Sub-Saharan Africa (Côte d’Ivoire) and Turkiye.

Life-course windows addressed today (records may span several stages)

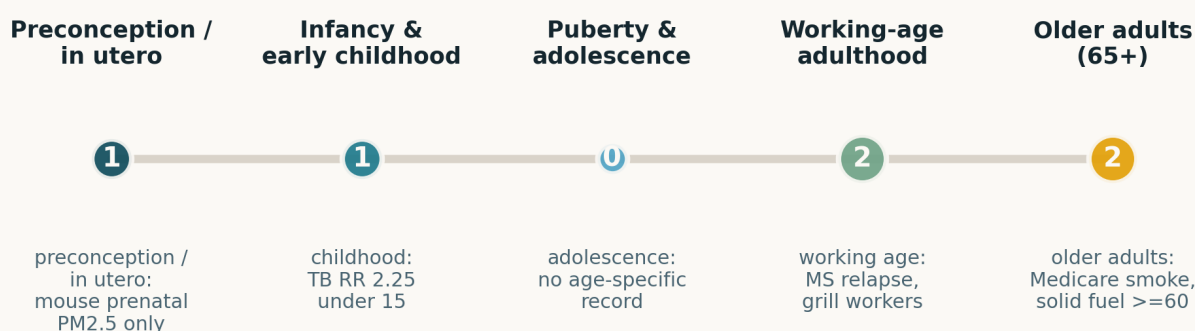


Fig. Life-course windows: in utero one (mouse only), childhood one (the TB under-15 RR), adolescence none, working age two (MS relapse, grill workers), older adults two (Medicare smoke, solid fuel ≥60).

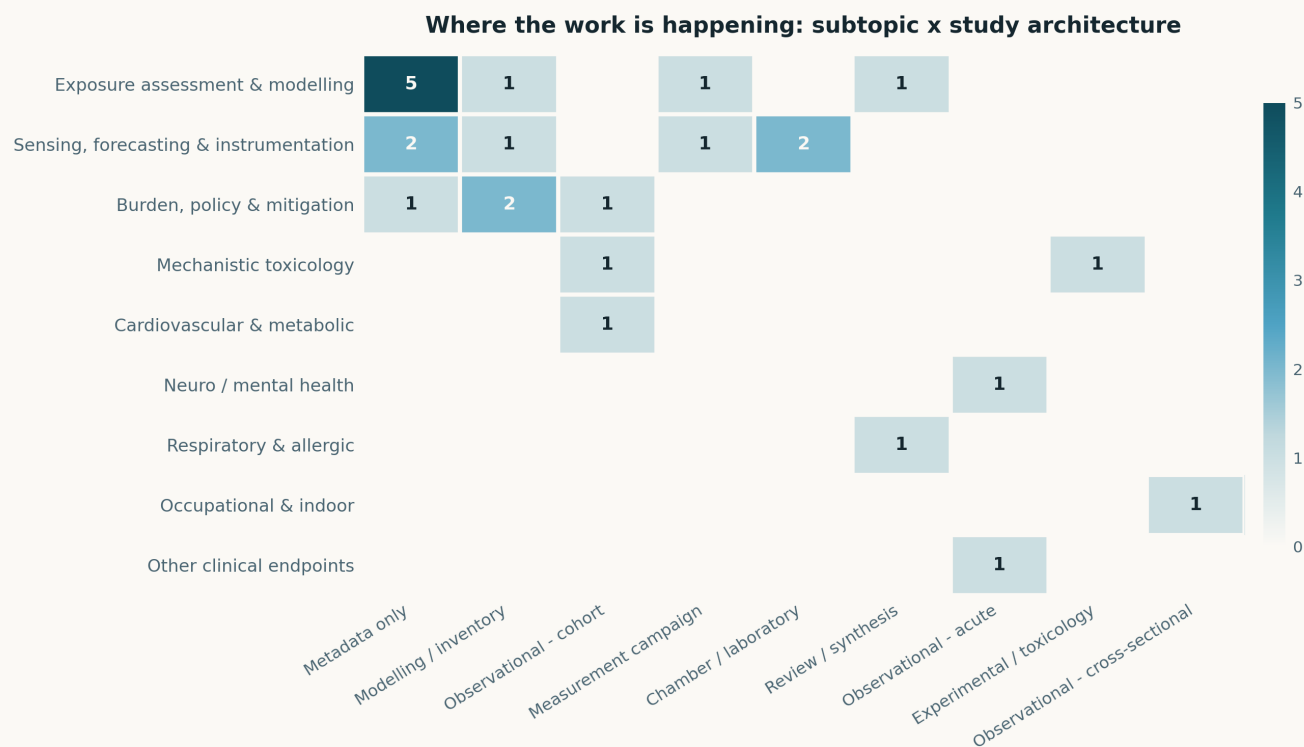


Fig. Subtopic × study architecture for the 25 records. The metadata-only column sits almost entirely in *Exposure assessment*; health designs are one-per-band, with no subtopic holding more than one epidemiological study today.



Fig. Nine ratio estimates with 95% CIs from six records, log scale. **Increments and contrasts differ** (per 1, per 10, per year of fuel use, fuel-type and occupational contrasts, and an unstated increment for the two MS rows), so the panel shows direction and precision, not a poolable effect. The cancer-mortality and smoke-mass rows are tight because of their size, not because the effect is large; the grill-worker dacrocyte row is the widest, from 25 workers per arm.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health

1 record

Januel et al. 2026 J Neurol Neurosurg Psychiatry

Air pollution and the risk of multiple sclerosis relapses in the Paris region, the CONFISEP study

1,152 adults with relapsing MS (898 relapses, 2018-2021) from the French MS registry; weekly Airparif street-level exposure at lags 0-15 weeks, analysed by self-controlled case series and cohort models adjusted for time-varying DMT, infections, temperature, sunlight and lockdown. PM₁₀ raised relapse risk at the 2-week (IRR 1.28, 95% CI 1.05-1.57) and 11-week lags (1.26, 1.02-1.54); PM_{2.5} at 2 weeks (1.20, 1.03-1.39); NO₂ at 2, 3 and 12 weeks. The exposure increment is not stated in the abstract, and with 16 lags x 4 pollutants the isolated 11-12-week hits are likely multiplicity, as the authors concede. DMTs dominated. Relevance: moderate - week-scale personal or neighbourhood sensing would sharpen exactly this lag structure.

doi: 10.1136/jnnp-2026-339534

Cardiovascular & metabolic

1 record

Sun et al. 2026 Indoor Air

Solid Fuel Use and Risks of All-Cause and Cardiovascular Disease Mortality in Urban Northern China: A 12-Year Cohort Study

Retrospective cohort of 39,054 urban residents in four northern Chinese cities followed 1998-2009: 1,435 non-accidental and 677 cardiovascular deaths. Each additional year of solid-fuel heating was associated with HR 1.15 (95% CI 1.14-1.17) for non-accidental, 1.15 (1.13-1.18) for cardiovascular and 1.16 (1.12-1.20) for cerebrovascular mortality; cooking HRs were slightly higher; adults ≥ 60 were most sensitive. A 15% hazard per year of use is implausibly steep for a cumulative-exposure metric and points to confounding by socioeconomic position, housing age and co-exposure; fuel-years is self-reported and no indoor PM was measured. Relevance: moderate - the exposure contrast is real, but quantifying it needs the indoor sensors this study lacks.

doi: 10.1155/ina/9645092

Respiratory & allergic

1 record

Carroll et al. 2026 Ann Am Thorac Soc

Global Tuberculosis Burden Attributable to Household and Outdoor Air Pollution

140 studies (to Aug 2024) pooled by random effects and Burden of Proof meta-regression. Household solid fuel: tuberculosis incidence RR 1.82 (95% CI 1.52-2.17), 2.25 in children and 2.62 in women. Short-term outdoor PM_{2.5}: RR 1.03 (1.00-1.05) per 10 $\mu\text{g m}^{-3}$. Attributed to PM_{2.5} in 2023: 4.15 million cases, 23.6 million DALYs and ~535,000 deaths - 49.9% of global TB. The attributable fraction is driven by the household-fuel RR, which comes from fuel-type contrasts confounded by poverty, crowding and undernutrition, and is far larger than the outdoor estimate whose lower bound touches 1. Treat the 50% as an upper bound. Relevance: moderate - household PM measurement is the missing exposure link.

doi: 10.1093/annalsats/aaog299

Mechanistic toxicology

2 records

Jiang et al. 2026 Redox Biol

Identification of shared and divergent molecular pathways linking ambient air pollutants to mortality and major chronic diseases through proteomic profiling

Olink plasma proteomics in 48,645 UK Biobank participants: proteins shared across PM_{2.5}, PM₁₀, NO₂ and NO_x were enriched in immune regulation; PM_{2.5}-specific proteins in vascular dysfunction, PM₁₀ in DNA damage, NO₂ in apoptosis, NO_x in matrix remodelling. Mediation and MR flag proteins causally linked to outcomes (vascular remodelling for CVD, oxidative stress for CKD, cell adhesion for dementia, AGER for chronic respiratory disease and lung cancer). No effect sizes or mediated proportions in the abstract. MR establishes protein-to-disease causality, not pollutant-to-protein, and the four pollutants come from the same 2010-era LUR surfaces, so 'pollutant-specific' pathways may be collinearity artefacts. Relevance: low-moderate for sensing.

doi: 10.1016/j.redox.2026.104411

Cai et al. 2026 Ecotoxicol Environ Saf

Prenatal PM_{2.5} exposure disrupts cortical excitatory/inhibitory balance in male offspring by suppressing Lhx8-regulated cholinergic homeostasis and cell adhesion

Male offspring of PM_{2.5}-exposed dams showed a persistent cortical shift toward inhibition (gephyrin up at PND 1, 7 and 21). RNA-seq and microarray identified 14 differentially expressed transcription factors; Lhx8 was consistently suppressed, with downstream ChAT loss, compensatory GABRA1 up-regulation and repression of adhesion genes (Cadm2, Cdh10, Pgrmc1, Cd36). Dose, route, PM source and litter-level analysis are not given in the abstract, males only, and no behavioural phenotype or Lhx8 rescue is reported, so causality within the proposed pathway is not tested. Relevance: low for sensing; supports the prenatal neurodevelopmental window seen in cohort data.

doi: 10.1016/j.ecoenv.2026.120862

Exposure assessment & modelling

8 records

Kuntz et al. 2026 J Hazard Mater*Dynamics of tire-related chemicals in roadside aerosol over one year in Leipzig, Germany-influencing factors, potential source materials and comparison with rural aerosol*

119 roadside PM₁₀ samples over 14 months were analysed for 40 tire-related chemicals. Medians: 6-PPD 0.61 pg/m³, 6-PPD-quinone 2.9 pg/m³, diphenylguanidine 90 pg/m³, benzothiazolesulfonic acid 41 pg/m³. Concentrations spanned about four orders of magnitude, with precipitation the dominant control, modulated by temperature, sunshine, wind and oxidants. Tread particles carried one to two orders of magnitude more TRC per mass than roadside aerosol, and rural levels were up to tenfold lower. Traffic-normalised concentrations are offered for extrapolation, but one roadside and one rural site cannot support transfer across fleets and climates. Relevance: moderate - a toxic tyre-wear fraction that optical sensors cannot distinguish within PM₁₀.

doi: 10.1016/j.jhazmat.2026.143749

Blond et al. 2026 Preprints.org (preprint)*Boundary-Layer Height Outweighs Wind Speed in the Meteorological Control of Urban Air Pollution: Unique-Variance Attribution and a City Typology for Ten French Cities*

Semi-partial R² separates the unique variance of boundary-layer height and wind speed for PM_{2.5}, PM₁₀, NO₂ and O₃ in France's ten largest cities (2018-2025), across pooled, deseasonalised and detrended frames, checked with GAMs. BLH and wind correlate at $r \sim 0.70$ (0.81 on anomalies); on pooled levels BLH carries four (NO₂) to nine (PM) times the unique variance of wind, a lead robust for PM. A three-type city typology (ventilation-, transitional, BLH-controlled) predicts which covariate a city needs. BLH is reanalysis-derived, not measured, and the preprint is not peer reviewed. Relevance: high for sensor-network trend normalisation and short-term epidemiology that adjust for wind alone - BLH should be a first-tier covariate for PM.

doi: 10.20944/preprints202609.2417.v1

Na et al. 2026 Atmos Environ*Satellite-based estimation of PM_{2.5} chemical composition under sparse observations in Seoul metropolitan area*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes satellite-based estimation of PM_{2.5} chemical components from sparse speciation observations. Relevant to satellite-surface fusion and to composition-specific health models. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122402

Rawat et al. 2026 Atmos Environ*Size-resolved aerosols from residential fuel combustion in the northwestern Himalayas: potential effects on light absorption and human health*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes size-resolved aerosol from household fuel combustion and its absorption and health implications. Relevant to household-air-pollution exposure, today's tuberculosis burden paper being the health side of the same source. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122399

Ma M. et al. 2026 Atmos Environ*Atmospheric Redistribution of Microplastics Reshapes Environmental Risk Across China*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes atmospheric transport of microplastics and its effect on risk across China. Airborne microplastics are part of the coarse fraction optical sensors mis-size. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122397

He B. et al. 2026 Atmos Environ*Contrasting seasonal behavior and inhalation risks of atmospheric PAHs, PAEs, and OPEs in a heating-influenced urban atmosphere*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes seasonal behaviour and inhalation risk of three semivolatile organic classes in a heating-season city. Composition data of this kind are what mass-only networks cannot supply. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122394

Niamien et al. 2026 Atmos Pollut Res*Spatio-temporal analysis of the physical properties of aerosols in urban and rural environments in Cote d'Ivoire*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes urban-rural aerosol physical properties in Cote d'Ivoire. West Africa is among the least monitored regions, where low-cost networks carry most of the observational load. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.apr.2026.103220

Guo et al. 2026 Atmosphere*Atmospheric Environment of the Northern Tianshan Urban Agglomeration, Xinjiang: A Comprehensive Review of Pollutant Characteristics, Photochemical Processes, and Health Implications*

Narrative synthesis of air-quality research in the Northern Tianshan agglomeration, grading evidence as local, regional Central Asian, or extrapolated. Aerosol shifts from dust-dominated coarse mode in spring to secondary inorganic and carbonaceous fine mode in winter; BC is local fossil combustion, BrC sources are speculative; ozone regime diagnoses are thin and there are no radical or oxidative-potential measurements. Local epidemiology is essentially absent, so the health section rests on extrapolation. No systematic search. Relevance: moderate - an arid, dust-coal mixture region where optical sensors need dust-aware correction and where a dense network would fill a documented gap.

doi: 10.3390/atmos17100961

Sensing, forecasting & instrumentation

6 records

Xu et al. 2026 Environ Sci Technol*Integrating Fragmented Mobile Monitoring Data into a High-Resolution Spatiotemporal Exposure Model for Black Carbon*

Vehicle-borne eBC measured in three seasonal campaigns (2023-2024) across residential Beijing, aggregated hourly, then gap-filled in time with XGBoost on external temporal predictors (regulatory pollutants, meteorology) and interpreted with SHAP; site-day estimates feed a hierarchical geostatistical model applied to pregnancy-cohort addresses. XGBoost reached 10-fold CV R² 0.69 against site-hour eBC; the spatial stage is reported only as internal interpolation at monitored points, so there is no external validation at unmonitored cohort homes - the step the cohort actually uses. Two stacked models also propagate error the paper does not quantify. Relevance: high - the same temporal-gap problem affects any campaign- or sparse-node design, and the two-stage template transfers directly to low-cost BC/PM networks.

doi: 10.1021/acs.est.6c08120

Herman et al. 2026 Atmos Environ*An Advanced, Multifunctional, and Extreme-Condition Test Chamber for Performance Evaluation of Air Quality Sensors*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes a chamber for evaluating air-quality sensors under extreme conditions. Directly relevant to LCS characterisation: humidity, temperature and high-load excursions are where field calibrations fail. First on the retry list. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.atmosenv.2026.122401

Mathissen et al. 2026 Atmosphere*Development of the Worldwide Harmonized Heavy Duty Brake Cycle (WHBC) for Brake Wear Particle Measurement*

Builds a regulatory brake-wear test cycle for heavy-duty vehicles (Euro 7 covers HDV non-exhaust PM but GTR 24 has only a light-duty cycle). Brake events were extracted from a global activity dataset and matched on empirical distributions of key brake metrics: 246 events over 271 km and 5 h 39 min. CO₂-monitoring data imply ~73% of kinetic energy is dissipated by friction brakes rather than retarders and drag. Dynamometer runs show the cycle is feasible and repeatable; particle results are described only as first insights, so no emission factor is reported. Limitation: retarder and regenerative braking share will shift with fleet electrification. Relevance: moderate - harmonised non-exhaust emission factors are what source-apportionment kerbside sensor signals.

doi: 10.3390/atmos17100964

Rosca & Stancu 2026 Indoor Air*Indirect Machine Learning Prediction of a Composite Indoor Air Quality Index in an Airport Passenger Transit Area*

A composite index (PIAQI) built from indoor CO₂, PM_{2.5} and PM₁₀ at Brindisi Airport is predicted without those three inputs, from other indoor aerosol channels, weather and public outdoor pollution data. Seven regressors were trained on 1 million high-frequency observations; XGBoost was best. Validation is a random 70/30 split of an autocorrelated series - the design Sahin et al. showed in the 29 Sep issue inflates R² - and the 'alternative aerosol measurements' retained are themselves close proxies for the excluded PM channels, so the task is closer to interpolation than to indirect sensing. Relevance: low - an example of the leakage-prone validation practice the field needs to retire.

doi: 10.1155/ina/3119205

Morel et al. 2026 Indoor Air*Evaluation of Two Cyclonic Bioaerosol Samplers for Hospital Air Monitoring in a Calibrated Test Chamber Integrating Aerodynamics and Biological Endpoints*

Coriolis+ and Coriolis Compact cyclones were challenged in a calibrated 20 m³ chamber with infectious SARS-CoV-2, S. aureus and A. fumigatus at several concentrations, scored on genomic and viability endpoints, and paired with CFD and airflow measurements. Neither detected infectious virus at the lowest level; Coriolis+ recovered nearly all viral genomes and more viable bacteria and fungi, while the Compact recovered more genomes of the two non-viral organisms at moderate-high loads. Ranking flips with endpoint, so 'collection efficiency' is not one number. No quantitative recoveries in the abstract. Relevance: low-moderate - bioaerosol, not PM mass, but a reference protocol for any sampler co-location.

doi: 10.1155/ina/2773156

Tu et al. 2026 Indoor Environ*Assessing the Need for Site-Specific Calibration of ATP Bioluminescence Sensors for Airborne Bacteria Monitoring in Indoor Environments*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title asks whether ATP-based airborne-bacteria sensors need site-specific calibration. The same transferability question PurpleAir regional-calibration work asks for optical PM sensors. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1016/j.indenv.2026.100197

Occupational & indoor

1 record

Yuksel & Yildirim 2026 Front Public Health*An analysis of occupational exposure to particulate matter and erythrocyte morphological abnormalities among charcoal-grill restaurant workers*

25 charcoal-grill restaurant workers vs 25 office staff. Kitchen PM median 117.8 vs 0.23 µg m⁻³ (the office value is implausibly low for any indoor space and suggests an instrument floor), with more wheeze (12/25 vs 0/25). After adjustment, four poikilocyte types were elevated - dacrococytes IRR 2.46 (95% CI 1.36-4.44), elliptocytes 2.33, sickle forms 4.47, schistocytes 1.56 - but total anomaly burden was not (IRR 1.57, 0.65-3.78). n=25 per arm, smears only for 19 controls, CO co-exposure unseparated and smoking unreported. Relevance: low-moderate - one more occupational combustion setting with nine-fold-plus PM gradients that cheap sensors could document over full shifts.

doi: 10.3389/fpubh.2026.1953388

Burden, policy & mitigation

4 records

Jin Z. et al. 2026 Environ Sci Technol*Carbonaceous Components Underlying Wildfire Smoke PM_{2.5}-Associated Mortality in Older U.S. Adults*

77.5 million Medicare beneficiaries (2002-2019), annual high-resolution smoke and non-smoke PM_{2.5} and carbonaceous matter, stratified Cox models. Per 1 $\mu\text{g m}^{-3}$: HR 1.032 (95% CI 1.027-1.037) for smoke PM_{2.5} and 1.087 (1.076-1.098) for smoke carbonaceous matter, both above the non-smoke counterparts per unit mass; per IQR the ranking reverses for mass because smoke exposure varies little. Estimated 22,420 deaths/yr attributable to smoke PM_{2.5}, mostly its carbon. Stronger in Black and dual-eligible beneficiaries. Annual means smooth episodic smoke into a chronic metric, and smoke attribution is itself modelled. Relevance: high - composition-resolved smoke exposure is the case for adding absorption/BC channels to LCS networks in fire regions.

doi: 10.1021/acs.est.6c07219

Manchanda et al. 2026 Environ Sci Technol*Inverse Modeling Identifies Efficient Emission Control Strategy for Mitigating PM_{2.5} Pollution*

Recasts attainment planning as a constrained Bayesian inverse problem: find the smallest spatially explicit, sector- and precursor-resolved emission changes that bring modelled PM_{2.5} under the 9 $\mu\text{g m}^{-3}$ annual NAAQS across the contiguous US. Strategies with similar total reductions ranged from full compliance to minimal progress depending on where and what was cut; aggregate tonnage does not predict attainment. No numeric results in the abstract; the solution inherits the transport model's linearised source-receptor relationships and inventory error. Relevance: moderate - attainment is judged at monitors, so where the monitors (and dense sensor networks) are determines which 'efficient' strategy appears to work.

doi: 10.1021/acs.est.6c06581

Ma et al. 2026 Environ Sci Technol*Afforestation and Emission Cuts: Dual Drivers for China's Clean Air and Carbon Neutrality*

Emission inventories plus a chemical transport model test maximum-potential afforestation against deep anthropogenic cuts. Afforestation raises summer BVOC emissions by 3.20-3.82 Tg (20-24%), adding 0.39 $\mu\text{g m}^{-3}$ PM_{2.5} (range 0.02-3.47) and 0.74 ppbv MDA8 O₃ in eastern China under current emissions; under carbon-neutral emissions SOA-related PM_{2.5} from afforestation falls 67% and the O₃ response turns slightly negative. Species choice alone can cut local O₃ impact to under a third. BVOC emission factors and SOA yields are the dominant, unquantified uncertainty. Relevance: low-moderate - the PM_{2.5} penalty is small relative to sensor error.

doi: 10.1021/acs.est.6c04722

Cheong et al. 2026 J Expo Sci Environ Epidemiol*Community-level air quality and health implications of sustainable aviation fuels adoption: a scoping review*

METADATA ONLY. Metadata-only record: Springer Nature deposited title, authors and DOI on 30 Sep with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes a scoping review of near-airport air quality and health under sustainable aviation fuel adoption. Relevant to ultrafine-particle monitoring around airports. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list.

doi: 10.1038/s41370-026-00971-7

Other clinical endpoints

1 record

Gyeltshen et al. 2026 Int J Cancer*Short-Term PM_{2.5} Exposure and Daily Cancer Mortality Across 47 Japanese Prefectures: A Nationwide Time-Stratified Case-Crossover Study, 2013-2022*

Nationwide case-crossover, all 47 prefectures 2013-2022, prefecture-specific conditional quasi-Poisson pooled by random effects. Per 10 $\mu\text{g m}^{-3}$ PM_{2.5} at lag 0-2, all-cancer mortality rose 0.88% (95% CI 0.57-1.18), with breast 2.06%, colorectal 1.06%, stomach 0.81% and lung 0.77%, and no effect modification by sex, age or site. The association faded beyond lag 0-5 and lost significance after adjustment for NO₂ or SO₂ - the decisive weakness: an acute cancer-death effect is most plausibly harvesting of frail terminal patients via cardiorespiratory pathways, and co-pollutant sensitivity leaves PM specificity open. Prefecture-mean exposure. Relevance: moderate - prefecture-scale exposure is coarse; finer networks would test the co-pollutant question.

doi: 10.1002/ijc.70761

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Januel et al.	Air pollution and the risk of multiple sclerosis relapses in the Paris region, the CONFISEP study	<i>J Neurol Neurosurg Psychia-</i> <i>try</i>	Self-controlled design	Weekly street-level PM ₁₀ , PM _{2.5} , NO ₂ , O ₃ (Airparif model)	France (Ile-de-France)
Cardiovascular & metabolic					
Sun et al.	Solid Fuel Use and Risks of All-Cause and Cardiovascular Disease Mortality in Urban Northern China: A 12-Year Cohort Study	<i>Indoor Air</i>	Retrospective cohort	Household solid fuel use for heating/cooking (years); no PM measured	China (Tianjin, Shenyang, Taiyuan, Rizhao)
Respiratory & allergic					
Carroll et al.	Global Tuberculosis Burden Attributable to Household and Outdoor Air Pollution	<i>Ann Am Thorac Soc</i>	Systematic review / meta-analysis	Household solid fuel; outdoor PM _{2.5} (short- and long-term)	Global / multi-region (review)
Mechanistic toxicology					
Cai et al.	Prenatal PM _{2.5} exposure disrupts cortical excitatory/inhibitory balance in male offspring by suppressing Lhx8-regulated cholinergic homeostasis and cell adhesion	<i>Ecotoxicol Environ Saf</i>	Animal experiment	Prenatal PM _{2.5} (mouse model; dose not in abstract)	Not stated (laboratory)
Jiang et al.	Identification of shared and divergent molecular pathways linking ambient air pollutants to mortality and major chronic diseases through proteomic profiling	<i>Redox Biol</i>	Cohort proteomics + mediation + Mendelian randomization	Residential PM _{2.5} , PM ₁₀ , NO ₂ , NO _x (LUR)	United Kingdom (UK Biobank)
Occupational & indoor					
Yüksel & Yildirim	An analysis of occupational exposure to particulate matter and erythrocyte morphological abnormalities among charcoal-grill restaurant workers	<i>Front Public Health</i>	Cross-sectional survey	Kitchen PM mass (spot), CO, CO ₂	Türkiye (Kocaeli)
Sensing, forecasting & instrumentation					
Herman et al.	An Advanced, Multifunctional, and Extreme-Condition Test Chamber for Performance Evaluation of Air Quality Sensors	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - sensor test chamber (by title)	Not recoverable from metadata
Mathissen et al.	Development of the Worldwide Harmonized Heavy Duty Brake Cycle (WHBC) for Brake Wear Particle Measurement	<i>Atmosphere</i>	Test-cycle development + dynamometer testing	Brake-wear particles (PM ₁₀ /PM _{2.5} , PN), heavy-duty vehicles	Global / multi-region (dataset)
Morel et al.	Evaluation of Two Cyclonic Bioaerosol Samplers for Hospital Air Monitoring in a Calibrated Test Chamber Integrating Aerodynamics and Biological Endpoints	<i>Indoor Air</i>	Laboratory sampler validation	Bioaerosols (SARS-CoV-2, <i>S. aureus</i> , <i>A. fumigatus</i>); cyclonic samplers	France (test chamber)
Rosca & Stancu	Indirect Machine Learning Prediction of a Composite Indoor Air Quality Index in an Airport Passenger Transit Area	<i>Indoor Air</i>	Machine-learning model	Indoor PM _{2.5} , PM ₁₀ and CO ₂ sensor data; composite IAQ index	Italy (Brindisi Airport)
Tu et al.	Assessing the Need for Site-Specific Calibration of ATP Bioluminescence Sensors for Airborne Bacteria Monitoring in Indoor Environments	<i>Indoor Environ</i>	Metadata only (no abstract)	Metadata only - ATP bioluminescence sensors for airborne bacteria (by title)	Not recoverable from metadata
Xu et al.	Integrating Fragmented Mobile Monitoring Data into a High-Resolution Spatiotemporal Exposure Model for Black Carbon	<i>Environ Sci Technol</i>	Mobile monitoring campaign	Equivalent black carbon (eBC), vehicle-platform aethalometry	China (Beijing)
Exposure assessment & modelling					
Blond et al.	Boundary-Layer Height Outweighs Wind Speed in the Meteorological Control of Urban Air Pollution: Unique-Variance Attribution and a City Typology for Ten French Cities	<i>Preprints.org (preprint)</i>	Statistical model	Station PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ vs boundary-layer height and wind	France (10 cities)
Guo et al.	Atmospheric Environment of the Northern Tianshan Urban Agglomeration, Xinjiang: A Comprehensive Review of Pollutant Characteristics, Photochemical Processes, and Health Implications	<i>Atmosphere</i>	Narrative review	PM _{2.5} , coarse dust, BC/BrC, O ₃ , VOCs (review)	China (Xinjiang)

First author	Focus	Journal	Design	Metric	Region
He B. et al.	Contrasting seasonal behavior and inhalation risks of atmospheric PAHs, PAEs, and OPEs in a heating-influenced urban atmosphere	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - particle/gas PAHs, phthalates, organophosphate esters (by title)	Not recoverable from metadata
Kuntz et al.	Dynamics of tire-related chemicals in roadside aerosol over one year in Leipzig, Germany-influencing factors, potential source materials and comparison with rural aerosol	<i>J Hazard Mater</i>	Measurement campaign	Roadside PM ₁₀ filters; 40 tire-related chemicals by LC-MS/MS	Germany (Leipzig)
Ma M. et al.	Atmospheric Redistribution of Microplastics Reshapes Environmental Risk Across China	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - airborne microplastics (by title)	China (national, by title)
Na et al.	Satellite-based estimation of PM _{2.5} chemical composition under sparse observations in Seoul metropolitan area	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - satellite-derived PM _{2.5} composition (by title)	South Korea (Seoul, by title)
Niamien et al.	Spatio-temporal analysis of the physical properties of aerosols in urban and rural environments in Cote d'Ivoire	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - aerosol physical properties (by title)	Cote d'Ivoire (by title)
Rawat et al.	Size-resolved aerosols from residential fuel combustion in the northwestern Himalayas: potential effects on light absorption and human health	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - size-resolved residential-combustion aerosol (by title)	India (north-western Himalayas, by title)
Burden, policy & mitigation					
Cheong et al.	Community-level air quality and health implications of sustainable aviation fuels adoption: a scoping review	<i>J Expo Sci Environ Epidemiol</i>	Metadata only (no abstract)	Metadata only - aviation-fuel emissions incl. ultrafine particles (by title)	Not recoverable from metadata
Jin Z. et al.	Carbonaceous Components Underlying Wildfire Smoke PM _{2.5} -Associated Mortality in Older U.S. Adults	<i>Environ Sci Technol</i>	Cohort	Annual smoke vs non-smoke PM _{2.5} mass and carbonaceous fraction	USA (contiguous)
Ma et al.	Afforestation and Emission Cuts: Dual Drivers for China's Clean Air and Carbon Neutrality	<i>Environ Sci Technol</i>	Chemical transport model	Summer PM _{2.5} (SOA) and MDA8 O ₃ from BVOC and anthropogenic scenarios	China (national)
Manchanda et al.	Inverse Modeling Identifies Efficient Emission Control Strategy for Mitigating PM _{2.5} Pollution	<i>Environ Sci Technol</i>	Chemical transport model	Annual PM _{2.5} vs precursor emissions by sector and location	USA (contiguous)
Other clinical endpoints					
Gyeltshen et al.	Short-Term PM _{2.5} Exposure and Daily Cancer Mortality Across 47 Japanese Prefectures: A Nationwide Time-Stratified Case-Crossover Study, 2013-2022	<i>Int J Cancer</i>	Time-stratified case-crossover	Daily prefecture PM _{2.5} (lag 0-2)	Japan (47 prefectures)

Provenance and method

Entry window **30 September 2026**, single day, continuous with the 29 September issue. Harvester legs run leg-by-leg: **PubMed** health 9, sensing 2; **Europe PMC** 39 (36 after the retro-index gate); **Crossref** by **ISSN 78**; **arXiv** no entry dated 30 September; **OpenAlex** HTTP 429. 125 raw deduplicated to 123, 13 already carried, 110 fresh. The **PubMed connector** ([EDAT] 30 September) returned 6 health and 0 sensing PMIDs, all already in the harvester set. 110 candidates screened, 25 admitted, 85 rejected with logged reasons (water chemistry and wastewater from *ES&T*, *Measurement Science and Technology* engineering, meteorology and carbon-cycle work from *Atmosphere*, gas-phase-only records, and a record with no DOI). **Consensus** returned 10 calibration hits, 0 new. **ClinicalTrials.gov**: one registry update (CAN-EXACT, NCT07850648, observational COPD-exacerbation cohort with environmental factors only as a secondary measure) screened, not added. **Scholar Gateway** still fails with an identity error. **Eight records carried on metadata alone**. DOI gate: 0 fail, 0 warn. Abstracts remain the copyright of their respective publishers.